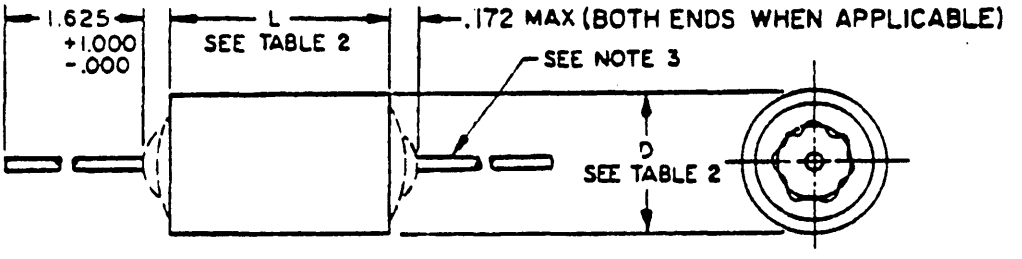
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR19

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER DATE OF THIS SPECIFICATION SHEET



INCHES	MM
.001	.03
.004	.10
.032	.81
.172	4.37
.332	8.43
.420	10.67
.562	14.27
1.000	25.40
1.562	39.67
1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Leads shall be of tinned solid wire, .020(No. 24 AWG) for cases .175 and .195 in diameter; .025(No. 22 AWG) for cases .235 thru .312 diameter: and .032(No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameter shall be +.004, -.001.
4. Capacitors with dimension L of 1.562 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be provided with supplementary means of mounting, such as a wrap-around band.

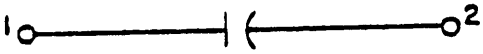
FIGURE 1. Style CQR19 capacitors.

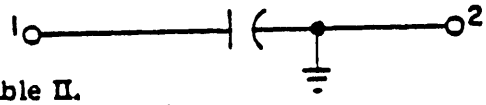
FSC 5910

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:05Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (end seal may be of magnetic material).
Dielectric material - Paper and polyethelene terephthalate (K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1  2

3  2

Rated voltage - See table II.
Rated temperature - -65° to +125°C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - See table II.
Dissipation factor (DF) (max) - 1.0 percent.
Failure rate level - In accordance with MIL-C-19978.
Reliability - In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter = A.

Dielectric withstanding voltage (DWV):
Sleeving - In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Sleeving - In accordance with MIL-C-19978.
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:05Z
Check the source to verify that this is the current version before use.

Salt spray: In accordance with MIL-C-19978.

Immersion:
DWV:
Sleeving - In accordance with MIL-C-19978.
IR:
Sleeving - In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

Moisture resistance:
DWV:
Sleeving - In accordance with MIL-C-19978.
IR:
Sleeving - In accordance with MIL-C-19978.

Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}{}^{+0^{\circ}}_{-5^{\circ}}$C.

Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}{}^{+5^{\circ}}_{-0^{\circ}}$C.

Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:05Z
Check the source to verify that this is the current version before use.

TABLE II. Style CQR19 capacitors.

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate level	Case dimensions			
					Circuit 1		Circuit 3	
					L±.031	D+.015 -.005	L±.031	D+.015 -.005
	Volts	µF			Inches	Inches	Inches	Inches
CQR19A-KD183-3-	300	.018	J, K	L,M,P,R,S	1.062	.332	.969	.332
CQR19A-KD223-3-	300	.022	J, K		1.062	.332	.969	.332
CQR19A-KD563-3-	300	.056	J, K		1.312	.420	1.219	.420
CQR19A-KD683-3-	300	.068	J, K		1.312	.420	1.219	.420
CQR19A-KD394-3-	300	.39	J, K		1.875	.770	1.812	.770
CQR19A-KD474-3-	300	.47	J, K		1.875	.770	1.812	.770
CQR19A-KD564-3-	300	.56	J, K		2.375	.770	2.312	.770
CQR19A-KD684-3-	300	.68	J, K		2.375	.770	2.312	.770
CQR19A-KD824-3-	300	.82	J, K		2.125	1.020	2.062	1.020
CQR19A-KD105-3-	300	1.0	J, K		2.125	1.020	2.062	1.020
CQR19A-KE184-3-	400	.18	J, K		1.500	.690	1.438	.690
CQR19A-KE224-3-	400	.22	J, K		1.500	.690	1.438	.690
CQR19A-KE566-3-	400	.56	J, K		2.625	.770	2.562	.770
CQR19A-KE684-3-	400	.68	J, K		2.625	.770	2.562	.770
CQR19A-KE824-3-	400	.82	J, K		2.250	1.020	2.188	1.020
CQR19A-KE105-3-	400	1.0	J, K		2.250	1.020	2.188	1.020
CQR19A-KF273-3-	600	.027	J, K		1.562	.420	1.469	.420
CQR19A-KF333-3-	600	.033	J, K		1.562	.420	1.469	.420
CQR19A-KF393-3-	600	.039	J, K		1.125	.582	1.062	.582
CQR19A-KF473-3-	600	.047	J, K		1.125	.582	1.062	.582
CQR19A-KF394-3-	600	.39	J, K		2.125	1.020	2.062	1.020
CQR19A-KF474-3-	600	.47	J, K	L,M,P,R,S	2.125	1.020	2.062	1.020

1/ Complete part number shall include symbols to indicate circuit, capacitance tolerance, and failure rate level, as applicable.

TABLE III. Millimeter equivalents of decimal inches.

Inch	mm	Inch	mm
.005	0.13	1.312	33.32
.015	0.38	1.438	36.53
.016	0.41	1.469	37.31
.031	0.79	1.500	38.10
.047	1.19	1.562	39.67
.062	1.57	1.812	46.02
.332	8.43	1.875	47.63
.420	10.67	2.062	52.37
.582	14.78	2.125	53.97
.690	17.53	2.188	55.58
.770	19.56	2.250	57.15
.969	24.61	2.312	58.72
1.020	25.91	2.375	60.33
1.062	26.97	2.562	65.07
1.125	28.57	2.625	66.67
1.219	30.96		

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T07:05Z
Check the source to verify that this is the currentversion before use.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU
Navy - AS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Air Force - 80

(Project 5910-1072-11)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:05Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

AFALD/PTR
Wright-Patterson AFB, OH 45433

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:05Z
Check the source to verify that this is the current version before use.